

- 3 A ball of mass M of 750 g is held on a smooth horizontal surface between two identical springs at their natural lengths as shown in Fig. 3.1.

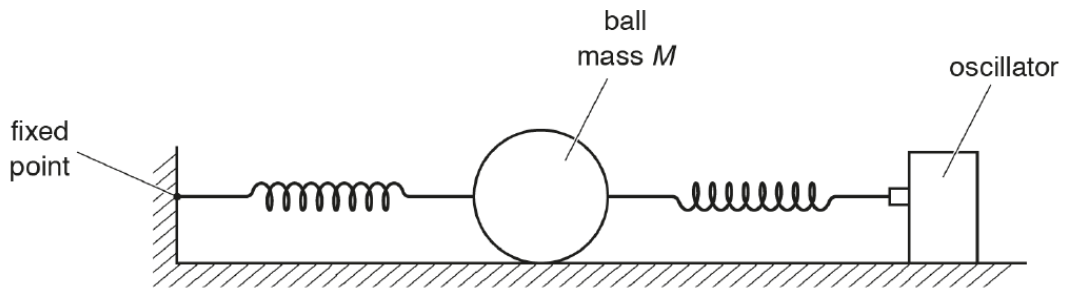


Fig. 3.1

One spring is attached to a fixed point while the other spring is attached to a mechanical oscillator. At $t = 0$ the ball is displaced to its amplitude position. The variation with time t of the displacement L of the ball is shown in Fig. 3.2.

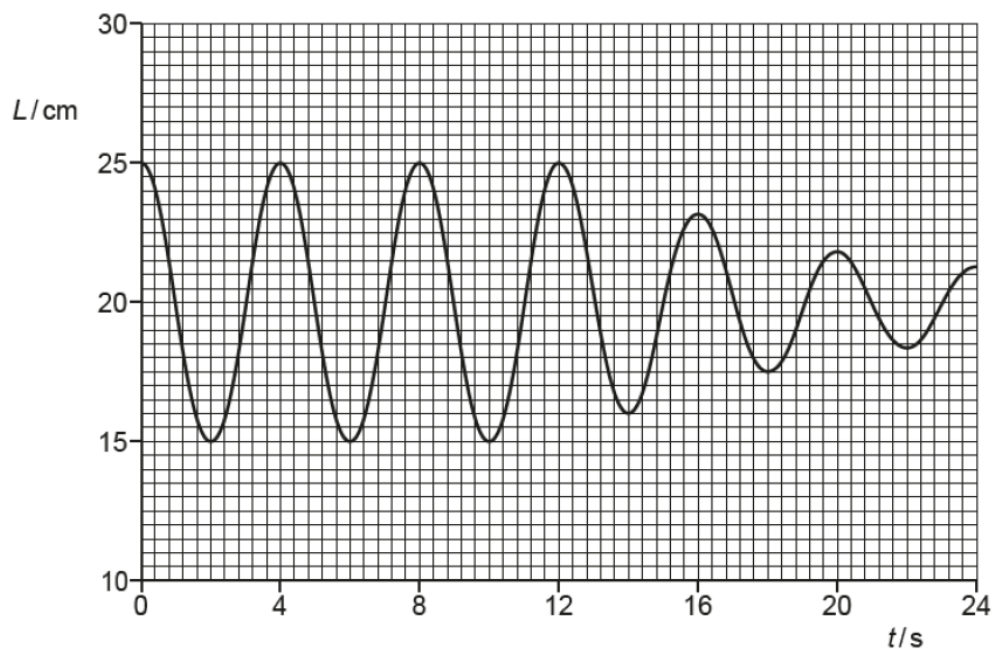


Fig. 3.2

- (a) For the first 12 s of the oscillations,

- (i) state one time at which the ball is moving with maximum speed,

Equilibrium: 1.0 s, 3.0 s, 5.0 s, 7.0 s, 9.0 s, 11.0 s.

time = s [1]

- (ii) state one time at which the springs have maximum elastic potential energy,

Max displacement: 0.0 s, 2.0 s, 4.0 s, 6.0 s, 8.0 s, 12.0 s.

time = s [1]

- (iii) calculate the angular frequency ω of the ball,

$$\omega = \frac{2\pi}{T} = \frac{2\pi}{4.0}$$

$$= 1.57 \text{ rad s}^{-1}$$

$\omega = \dots\dots\dots \text{ rad s}^{-1}$ [1]

(iv) calculate the maximum acceleration of the ball.

$$\begin{aligned}
 a_{\max} &= \omega^2 x_0 \\
 &= (1.57)^2 (0.05) \\
 &= 0.123 \text{ m s}^{-2}
 \end{aligned}$$

maximum acceleration = m s⁻² [2]

(b) Some salt is sprinkled on the horizontal surface at $t = 12.0$ s.

Calculate the loss in total energy of the oscillations during the first 24 s of the oscillations.

Show your working clearly.

At $t = 0$ s, $x_0 = 0.05$ m

$$\begin{aligned}
 \text{Total } E &= \frac{1}{2} m v_{\max}^2 = \frac{1}{2} m \omega^2 x_0^2 \\
 &= \frac{1}{2} (0.750) (1.5708)^2 (0.05)^2 \\
 &= 2.3132 \times 10^{-3} \text{ J}
 \end{aligned}$$

At $t = 24$ s, $x_0 = 0.0125$ m

$$\begin{aligned}
 \text{Total } E &= \frac{1}{2} m v_{\max}^2 = \frac{1}{2} m \omega^2 x_0^2 \\
 &= \frac{1}{2} (0.750) (1.5708)^2 (0.0125)^2 \\
 &= 1.4457 \times 10^{-4} \text{ J}
 \end{aligned}$$

Loss in Total E = 2.17×10^{-3} J

[3]

[Total: 8]

